

Solving MDPs: Value Iteration and Policy Iteration

Bellman optimality, contraction, greedy improvement, and convergence

Learning objectives

By the end of this lesson, you should be able to:

- 01 — Distinguish Bellman expectation from Bellman optimality.
- 02 — Perform a value-iteration backup.
- 03 — Explain the gamma-contraction convergence intuition.
- 04 — Extract a greedy policy from values.
- 05 — Trace policy evaluation and policy improvement.
- 06 — Compare value iteration with policy iteration.

Optimal control replaces policy averaging with maximization

The Bellman optimality operator chooses the action with the greatest expected one-step reward plus continuation value.

- Expectation evaluates a specified policy.
- Optimality maximizes over actions.
- A greedy policy with respect to V^* is optimal.

Fixed point

V^* is unchanged by another optimality backup.

Expectation and optimality use different backups

$$T^{\pi} V: \sum_a \pi(a|s) q_V(s,a) \qquad T^* V: \max_a q_V(s,a)$$

- q_V includes P , R , and γV .
- T^{π} keeps policy fixed.
- T^* selects the best action at each state.
- $V^* = T^* V^*$.

ENGINEERING IMPLICATION

Do not average actions when computing an optimality backup.

Value iteration mixes evaluation and improvement

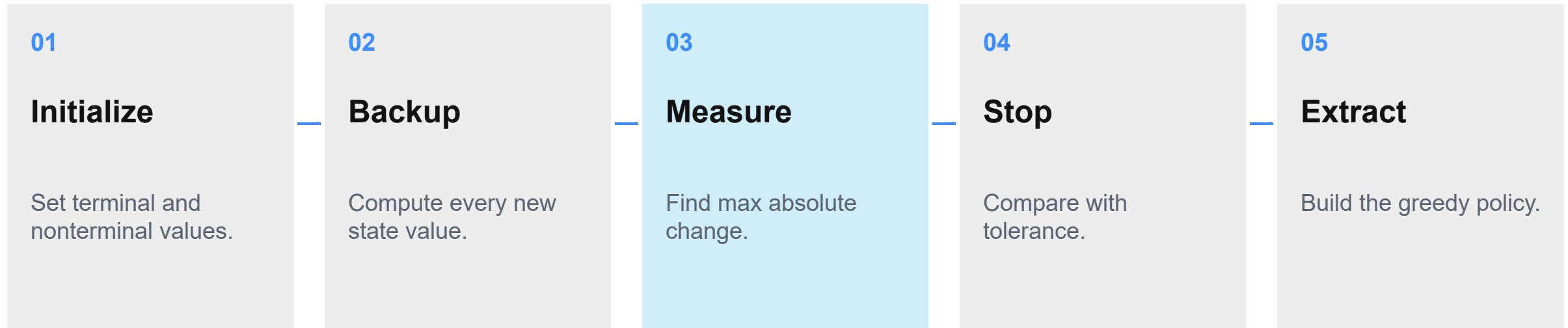
Value backup

- Initialize V arbitrarily.
- Apply T^* to every nonterminal state.
- Use a synchronous copy for clear semantics.
- Stop by max-norm residual.

Policy extraction

- Compute $q_V(s,a)$.
- Choose an argmax action.
- Break ties deterministically.
- Terminal states have no action.

Value iteration converges through repeated contraction



A stopping rule should measure a documented residual, not an arbitrary iteration count alone.

Value iteration uses a synchronous Bellman sweep

```
repeat:
    delta = 0
    newV = copy(V)
    for s in nonterminal_states:
        candidates = []
        for a in actions(s):
            candidates.append(expected_backup(s, a, V))
        newV[s] = max(candidates)
        delta = max(delta, abs(newV[s]-V[s]))
    V = newV
until delta < tolerance
```

Implementation rules

- Terminal value is fixed.
- Read old V in one sweep.
- Normalize transitions.
- Guard max iterations.

One value-iteration backup

SCENARIO

State A has actions Exit for reward 2, or Wait for reward 0 then return to A; $\gamma=0.9$ and current $V(A)=5$.

- $Q(A, \text{Exit})=2$.
- $Q(A, \text{Wait})=0+0.9*5=4.5$.
- Take the maximum.
- Set new $V(A)=4.5$.

The greedy action under the current values is Wait, even though later sweeps may change it.

Contraction turns approximation into a convergence guarantee

Operator

Bellman optimality

- Maps one value vector to another.
- Uses max over actions.
- Preserves bounded values.
- Has fixed point V^* .

Contraction

Max norm

- Distance shrinks by at most gamma.
- Requires $\gamma < 1$.
- Repeated backups converge.
- Residual bounds error.

Tolerance

Finite stopping

- Tracks max change.
- Balances time and precision.
- Depends on gamma.
- Report iterations.

Checkpoint: compute an optimality backup

CHECKPOINT

Action a has expected backup 4.2 and action b has expected backup 3.7. What value and greedy action result?

01

Identify the optimality operator.

02

Take the maximum.

03

State the tie rule if values were equal.

Policy iteration separates evaluation and improvement

Policy evaluation

- Hold π fixed.
- Solve or iterate $T^{\pi} V = V$.
- Estimate V^{π} .
- Stop at evaluation tolerance.

Policy improvement

- Hold V fixed.
- Choose $\arg\max_a q_V(s, a)$.
- Replace inferior actions.
- Stop when policy is stable.

Greedy improvement cannot reduce policy value

$$\pi'(s) \text{ in } \operatorname{argmax}_a q^{\pi}(s,a) \text{ implies } V^{\pi'}(s) \geq V^{\pi}(s)$$

- q^{π} evaluates one deviation then follows π .
- π' chooses a maximizing deviation.
- Policy improvement holds statewise.
- A stable greedy policy satisfies optimality.

ENGINEERING IMPLICATION

Finite deterministic policies prevent endless strict improvement.

One policy-improvement step

SCENARIO

Under current V^{π} , state A's action values are Left=1.2, Right=2.8, and Stay=2.1.

- Evaluate all legal actions with the same V^{π} .
- Find the maximum value 2.8.
- Set $\pi'(A)=\text{Right}$.
- Record whether the action changed.

Right replaces the old action unless it was already selected; stability is checked across all states.

Value and policy iteration trade sweep cost for count

Value iteration

Cheap mixed sweeps

- One optimality sweep per iteration.
- Many simple iterations.
- Easy vectorization.
- Policy extracted at end.

Policy iteration

Evaluate then improve

- Fewer outer iterations.
- Evaluation can be expensive.
- Policy is explicit throughout.
- Exact or approximate evaluation.

Modified PI

Intermediate option

- Partial evaluation.
- Regular improvement.
- Balances costs.
- Needs clear stopping rules.

Checkpoint: diagnose a terminal-state bug

CHECKPOINT

A solver applies gamma $V(\text{terminal})$ after entering a terminal state and repeatedly collects its reward. What is wrong?

01

State the intended terminal semantics.

02

Identify double counting.

03

Describe the code-level fix.

Numerical convergence needs observable evidence

Report residual, iteration count, policy stability, and validation against hand-solvable MDPs.

- Use max-norm residual and a maximum-iteration guard.
- Compare VI and PI policies on the same model.
- Test $\gamma=0$, γ near 1, terminal-only, and unreachable states.

Convergence

A loop ending is not evidence that the intended fixed point was reached.

Key takeaways

- 01 — Bellman optimality maximizes rather than averages policy actions.
- 02 — Value iteration converges because T^* is a gamma-contraction.
- 03 — Policy iteration alternates evaluation and greedy improvement.
- 04 — Terminal semantics and tolerance determine reliable implementations.

Next: Week 5 removes the known transition model and learns values from sampled interaction.

Check your understanding

Q1

Compute one value-iteration sweep.

Q2

Explain the contraction intuition.

Q3

Extract a greedy policy.

Q4

Trace one policy-improvement step.

Q5

Compare VI and PI on per-iteration cost.